



Instructor: Date:

Learner: Regional Branch:

Program: Center:

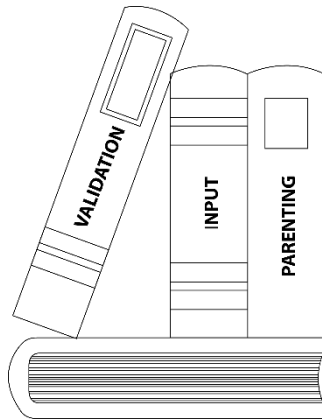
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Let's Start!

Databases and programming!



Learning Outcome:

Exchange written and spoken information on principles, concepts, data base objects related to relational database management system in spoken or written texts.
Describe relational database, its concepts, and principles in spoken or written texts.

Can Do Statements:

- I can understand written information about the SQL programming language for relational databases.
- I can identify some commands to use in the SQL programming language.
- I can write a simple text to talk about relational databases and the SQL programming language.

Ask your partner:

- What can you mention about the relationship between databases and programmers?
- What do you think the most difficult element of working with databases in the engineering field is?



Let's Explore!

Two friends study for a SQL exam.



Instructions: Read the text below.

Rob: Thank you for tutoring me for the exam, Laila.

Laila: No problem! Pay lots of attention and you will get an excellent grade. Now, I presented that exam two semesters ago, so I know that the exam is going to include many questions about SQL.

Rob: SQL?

Laila: Yes. You are going to have to remember that SQL is the programming language that we use in relational databases. Furthermore, you will have to keep in mind that it contains manipulation languages as well as data definition.

Rob: Wait, could it be used for anything?

Laila: Of course! SQL may be perceived as of great help for today's engineering field. In fact, any programmer will have to use it to design, modify and edit database schema.

Rob: Wow, I have a question. How do they use SQL?

Laila: SQL is used by programmers through different types of commands.

Rob: So, should commands be studied for the exam too?

Laila: Sure, commands can be seen as the most important element of this topic. Commands can be implemented in the databases to create, drop or alter information. Plus, you will have to use other sets of commands to select, insert, update or delete values within the databases. Let's check thoroughly some of those. For example, we can find the command "Select author_name From article_author Where nationality>French" to have a database select an element with those characteristics. On the other hand, we can find "DELETE FROM table_name" too with which we want to completely remove elements from a table. Finally, if we want to create something like a table in the database, we will find a command such as "Create table;".





Let's Practice!

What can we do with those commands?

Instructions: Use the words from the box to correctly complete the sentences. Read the sentences aloud. Check your pronunciation with your instructor.

Word Bank

remove elements / can be seen / to alter information /
can be used / used to create information

1. Commands _____ as important elements of SQL.
2. These commands should be _____.
3. "Select author_name From article_author Where nationality>French"
_____ to select an element with similar characteristics.
4. Commands can be used if you want _____.
5. "DELETE FROM table_name" is used to _____.



Let's Practice!

Laila is telling Rob what he will have to do when working with SQL.

Instructions: Match the sentences from Column A with Column B. Check your answers with your partner. Discuss.

COLUMN A

1. You are going to use
2. You will use SQL in
3. You are going
4. You will have to remember that
5. You will use SQL

COLUMN B

- a. SQL is a programming language.
- b. through different commands.
- c. SQL if you work as a programmer.
- d. relational databases.
- e. to use SQL to design database schema.



Extension Activity

Let's find out more about SQL commands!

Instructions: Complete the following steps. Fill in the Self-Assessment.

- Look for more commands used in SQL.
- Make a list of the additional commands you found and also their functions.
- Share your list with a partner and explain why those commands are useful.

Self-Assessment

Yes

No

- I identify more commands for SQL.
- I list commands and their uses.
- I discuss my list with a partner

☐	☐
☐	☐
☐	☐



Target Words

- **relational database** (noun): a type of database that organizes information in tables
e.g. You are going to have to remember that SQL is the programming language that we use in relational databases.
- **manipulation language** (noun): A series of elements that we can use to insert or modify data in a database
e.g. You are going to have to remember that SQL is the programming language that we use in relational databases.
- **check thoroughly** (expression): to review something in a complete way
e.g. Let's check thoroughly some of those.

- **tutor** (verb): to teach something to someone in an extra session
e.g. Thank you for tutoring me for the exam, Laila.
- **drop** (verb): to erase something
e.g. Commands can be implemented in the databases to create, drop or alter information.
- **alter** (verb): to modify or change something
e.g. Commands can be implemented in the databases to create, drop or alter information.



Learning Points

Passive Voice

You use the **passive voice** to put the focus of the sentence on its **object** rather than on its **subject**.

A sentence that follows the structure *Subject + verb + object + complement* gives importance to the **doer**, the **subject**, of the sentence. Now, if what is really important is the **object** of the sentence, we can make some changes on the structure that we follow, leaving behind the **doer**.

Object + verb to be (am /is /are) + action verb (past participle form) + doer (which is optional and introduced with "by")

Examples:

*Software programming **is designed** by programmers from all over the world. (We mention the doer)*

*Faster computers **are created** year after year. (Omission of doer)*

Now, we can also include **modal verbs** within sentences with the **passive voice**. For affirmative sentences, follow the structure:

Object + may/ must/ can/ could/ ought to/ should + be + verb (past participle form)

For questions, follow the structure:

May/ Must/ Can/ Could/ Ought to/ Should + object + be + verb (past participle form) + ?

Example:

*Should software be free for all users? Software **can be sold** for free in a few years.*

Examples

- *SQL may be perceived as of great help for today's engineering field.*
- *So, should commands be studied for the exam too?*



Learning Points

Future Simple, Review

The **future simple** allows you to express, in different ways, ideas about the future.

There are two different expressions you can use to talk about the **future**. Each one of them implies a different meaning:

Will

Remember that we can use **will** to talk about decisions that we take right at the moment of speaking. We can also use it to make promises and unlikely predictions. Furthermore, expressions like **probably, perhaps, maybe**, among others can be used for these purposes. The structure that we use is the following:

Example:

*The company **will** probably launch a new product this year. I **will** call the company.*

Be going to

Don't forget that we can use **be going to** to talk about plans in the future. These are planned before the moment of speaking. We can also use it to talk about predictions that we make based on evidence.

The structure that we follow is:

Subject + verb to be (am/is/are) + going to + verb (base form) + complement.

Example:

*That computer **is going to** explode, look at all the smoke.*

*Next month we **are going to** travel to Japan to learn more about software.*

Examples

- *Now, I presented that exam two semesters ago so I know that the exam is going to include many questions about SQL.*
- *Furthermore, you will have to keep in mind that it contains manipulation languages as well as data definition.*